

Cloud Computing & Microsoft Azure

1. What is Cloud Computing?

Cloud computing is the delivery of computing services — servers, storage, databases, networking, software, and more — over the internet ("the cloud"). Instead of owning physical hardware, you rent what you need from a provider and pay only for what you use.

Key points:

- No need to buy or maintain physical servers.
- Access resources from anywhere with an internet connection.
- Scale up or down instantly depending on your needs.
- Pay-as-you-go pricing — no upfront cost.

2. What are the Advantages of Microsoft Azure?

Azure is Microsoft's cloud platform. Here's why companies use it:

- **Global reach:** Data centers in 60+ regions worldwide.
- **Security:** Built-in compliance tools and enterprise-grade security.
- **Integration:** Works seamlessly with Microsoft products like Office 365, Teams, and Windows Server.
- **Scalability:** Instantly scale resources up or down based on demand.
- **Cost-effective:** Pay only for what you use; no hardware investment needed.
- **Hybrid support:** Easily connect on-premises infrastructure with the cloud.

3. What is Hosting in Azure?

Hosting in Azure means running your applications, websites, or databases on Azure's infrastructure instead of your own servers.

Common hosting options in Azure:

- **Azure App Service:** Host web apps and APIs without managing the underlying server.
- **Azure Virtual Machines (VMs):** Full control over an OS and environment, just like a physical server.
- **Azure Kubernetes Service (AKS):** Host containerized applications at scale.
- **Azure Static Web Apps:** Host simple front-end websites quickly and cheaply.

In short: you push your code or app to Azure, and Azure takes care of running it reliably.

4. What is IaaS, PaaS and SaaS?

These are the three main cloud service models. Think of them as different levels of control vs. convenience.

IaaS – Infrastructure as a Service

You get the raw infrastructure: virtual machines, storage, networking. You manage the OS, software, and everything above it. The provider only maintains the physical hardware.

- **Example:** Azure Virtual Machines
- **Best for:** Full control, custom environments, lift-and-shift migrations.

PaaS – Platform as a Service

You get a platform to build and deploy applications. The provider manages the OS, runtime, and infrastructure. You only focus on your code.

- **Example:** Azure App Service, Azure SQL Database
- **Best for:** Developers who want to build apps without worrying about servers.

SaaS – Software as a Service

You use a fully built application delivered over the internet. No installation or maintenance needed — just log in and use it.

- **Example:** Microsoft 365, Outlook, Teams, Salesforce
- **Best for:** End users who just need to use software without touching any infrastructure.

Quick Comparison:

- IaaS → You manage: OS, apps, data. Provider manages: hardware.
- PaaS → You manage: apps, data. Provider manages: OS + hardware.
- SaaS → You manage: nothing. Provider manages: everything.

5. What is EDA – Exploratory Data Analysis?

EDA is the process of examining a dataset before doing any formal analysis or modeling. The goal is to understand what the data looks like, spot patterns, catch errors, and figure out what questions you can answer with it.

What you typically do in EDA:

- **Check the shape of the data:** How many rows and columns? What data types are there?
- **Handle missing values:** Find nulls or blanks and decide whether to fill or drop them.
- **Summary statistics:** Mean, median, min, max, standard deviation — get a feel for the numbers.
- **Spot outliers:** Values that are unusually high or low and might affect your analysis.
- **Visualize distributions:** Use histograms, box plots, and bar charts to see how data is spread.
- **Find relationships:** Check correlations between columns to see what variables are related.

Common tools used: Python (Pandas, Matplotlib, Seaborn), SQL, Power BI, Excel.

Why it matters: Skipping EDA often leads to bad models and wrong conclusions. It's the first thing any data engineer or analyst should do before touching the data.